

4 – Execute the Burgwald Data Pipeline

Clone · Configure · Run · Verify

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Inhaltsverzeichnis

Goals	1
Repository	2
Task 1 – Basic project layout	2
Task 2 – Run the setup	2
Task 3 – Run generic data retrieval	3
Task 4 – Test external API (SAGA)	3
Task 5 – Sentinel-2 via CDSE	4
Task 6 – Sentinel-2 via gdalcubes	4
Task 7 – Pipeline check	5
Task 8 – Short reproducibility log	5
Expected final structure	5

Goals

- Clone the course repository and open the RStudio project.
- Run all core pipeline scripts **unchanged** on your own machine.
- Verify that raw data, processed data, and predictor stacks are created.
- Document any problems that prevent full reproducibility.

Repository

Clone this repository locally and open the .Rproj:

<https://github.com/gisma/LV-19-d19-006-25-R.git>

Task 1 – Basic project layout

After cloning and opening the project, check that these top-level folders exist (create if missing):

```
data/raw/  
data/processed/  
data/predictor/  
src/  
outputs/
```

Task 2 – Run the setup

Script: 01_setup-burgwald.R

- Open the script.
- Run it from top to bottom **without any modifications**.
- After successful execution, the following must exist:

```
data/raw/AOI_Burgwald/  
data/processed/aoi_burgwald.gpkg  
src/01-fun-data-retrieval.R (loaded without error)
```

Minimal check (optional in console):

```
fs::dir_ls("data", recurse = TRUE)
```

Task 3 – Run generic data retrieval

Script: 01-data-retrieval.R

- Run the script unchanged.
- Afterwards, the following data must physically exist (content not relevant, existence is):
 - DEM (GeoTIFF)
 - OSM-by-key files
 - CLC raster / GeoPackage
 - DWD station data in `data/raw/dwd-stations/`
 - processed DWD files in `data/processed/dwd/`

Suggested quick check:

```
fs::dir_ls("data", recurse = TRUE)
```

Task 4 – Test external API (SAGA)

Script: 01-04-1-SAGA.R

- Adjust the SAGA binary path if needed (only this, nothing else).
 - Run the script.
 - At least **one new terrain raster** (e.g. slope, aspect) has to be created.
 - Note the file name and location.
-

Task 5 – Sentinel-2 via CDSE

Script: 01-2-CDSE-sentinel-data-retrieval.R

- Run the script unchanged.
- Expected result (folder may vary slightly, adapt to repo):

```
data/raw/s2/  
  <Sentinel-2 scenes or COG references>
```

- Count how many items were downloaded or referenced.
 - Check if a STAC JSON file was created (if implemented in the script).
-

Task 6 – Sentinel-2 via gdalcubes

Script: 01-3-gdalcubes-sentinel-data-retrieval.R

- Run the script unchanged.
- Expected output:

```
data/predictor/  
  <NetCDF file>    # e.g. s2_predictor_stack_summer.nc
```

- Confirm that the NetCDF file exists and can be opened, e.g.:

```
stars::read_stars("data/predictor/<your-file>.nc")
```

Task 7 – Pipeline check

Run this final check:

```
list.files("data/raw", recurse = TRUE)
list.files("data/processed", recurse = TRUE)
list.files("data/predictor", recurse = TRUE)
```

Verify that all three stages exist on your system:

- RAW DATA
- PROCESSED DATA
- PREDICTOR STACK(S)

If any stage is missing, re-run the corresponding script or document the error.

Task 8 – Short reproducibility log

Write **one short paragraph** (max. 5–6 sentences) for your own log:

- Did the full pipeline run on your machine?
- Which scripts worked without issues?
- Where did you have to change paths or credentials (e.g. SAGA, CDSE)?
- Which errors (if any) stopped the pipeline?

You will use this later to compare reproducibility between systems.

Expected final structure

The expected directory tree at the end of this worksheet is:

```
data/  
  raw/  
    AOI_Burgwald/  
    dem/  
    osm_by_key/  
    dwd-stations/  
    s2/  
  processed/  
    aoi_burgwald.gpkg  
    dwd/  
  predictor/  
    *.nc  
src/  
  setup/  
  retrieval/  
  remote_sensing/  
outputs/  
  figures/  
  tables/
```
